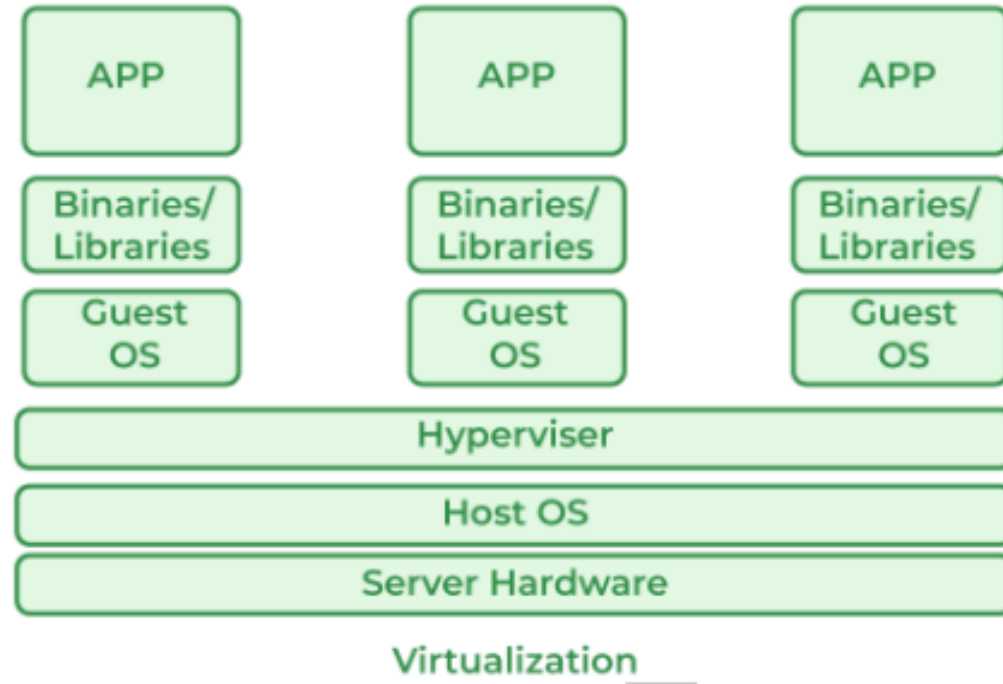


Virtualization is used to create a virtual version of an underlying service. With the help of Virtualization, multiple operating systems and applications can run on the same machine and its same hardware at the same time, increasing the utilization and flexibility of hardware. It was initially developed during the mainframe era.

It is one of the main cost-effective, hardware-reducing, and energy-saving techniques used by cloud providers. Virtualization allows sharing of a single physical instance of a resource or an application among multiple customers and organizations at one time. It does this by assigning a logical name to physical storage and providing a pointer to that physical resource on demand.



The term virtualization is often synonymous with hardware virtualization, which plays a fundamental role in efficiently delivering Infrastructure-as-a-Service (IaaS) solutions for [cloud computing](#). Moreover, virtualization technologies provide a virtual environment for not only executing applications but also for storage, memory, and networking

- **Host Machine:** The machine on which the virtual machine is going to be built is known as Host Machine.
- **Guest Machine:** The virtual machine is referred to as a Guest Machine.

Working of Virtualization in Cloud Computing

Virtualization has a prominent impact on Cloud Computing. In the case of cloud computing, users store data in the cloud, but with the help of Virtualization, users have the extra benefit of sharing the infrastructure.

Cloud Vendors take care of the required physical resources, but these cloud providers charge a huge amount for these services which impacts every user or organization. Virtualization helps Users or Organisations in maintaining those services which are required by a company through external (third-party) people, which helps in reducing costs to the company. This is the way through which Virtualization works in Cloud Computing.

What is a Secure SDLC (SSDLC)?

The Secure Software Development Lifecycle (SSDLC) is a critical framework that integrates security measures into every phase of the software development process. By embedding security from the initial design through to deployment, SSDLC ensures that potential vulnerabilities are addressed proactively. This approach not only enhances the overall security posture of applications but also reduces the risk of breaches and data loss. Examples include designing applications to ensure that your architecture will be secure, as well as including security risk factors as part of the initial planning phase.

SLDC phases

[Software Development Lifecycle \(SDLC\)](#) describes how software applications are built. It usually contains the following phases:

- **Requirements** gathering
- **Analysis** of the requirements to guide design
- **Design** of new features based on the requirements
- **Development** of new capabilities (writing code to meet requirements)
- **Testing and verification** of new capabilities—confirming that they do indeed meet the requirements
- **Deployment** of the new project

Software Development Life Cycle (SDLC)



What is Application Security?

Application security aims to protect software application code and data against cyber threats. You can and should apply application security during all phases of development, including design, development, and deployment.

Here are several ways to promote application security throughout the software development lifecycle (SDLC):

- Introduce security standards and tools during design and application development phases. For example, include vulnerability scanning during early development.
- Implement security procedures and systems to protect applications in production environments. For example, perform continuous security testing.
- Implement strong authentication for applications that contain sensitive data or are mission critical.



What is Web Application Security?

Web application security is the practice of protecting websites, applications, and APIs from attacks. It is a broad discipline, but its ultimate aims are keeping web applications functioning smoothly and protecting business from cyber vandalism, data theft, unethical competition, and other negative consequences.

The global nature of the Internet exposes web applications and APIs to attacks from many locations and various levels of scale and complexity. As such, web application security encompasses a variety of strategies and covers many parts of the software supply chain.

What is Remote Access Security?

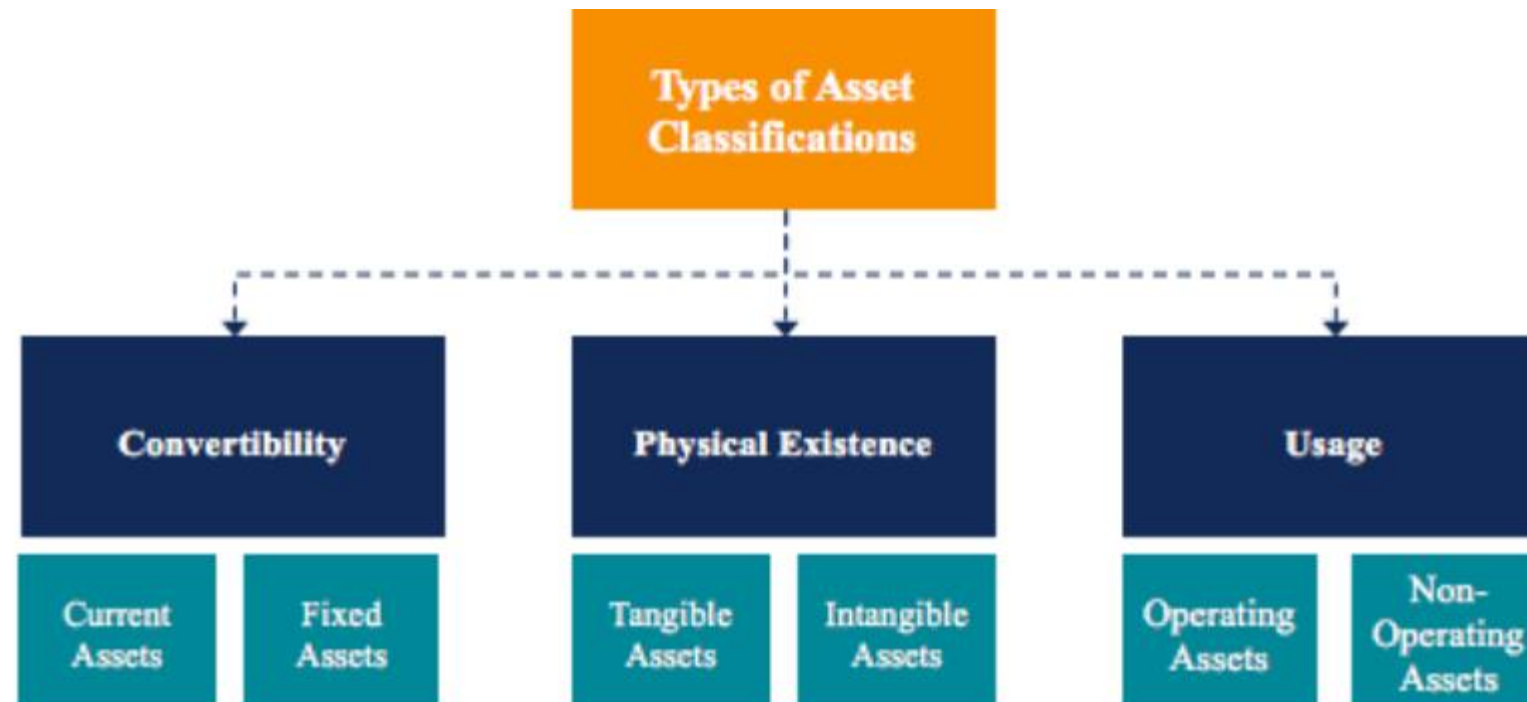
Organizations use [remote access security](#) solutions to authenticate users who are accessing business applications and IT systems from outside the private enterprise network. Today's business users are no longer confined to the office. To be fully productive, they need secure and convenient access to all their on-premises and cloud-based applications and systems when working from home or the road. For cybersecurity leaders, it's important to understand which remote access security technologies can enable employees to succeed, while protecting the organization from attackers.

Remote access security solutions allow organizations to safely extend business applications and services to teleworkers and nomadic users without impairing user experience or productivity. Most contemporary remote access security solutions support [Multi-Factor Authentication](#) (MFA) functionality to validate a remote user's identity and [Single Sign-On \(SSO\)](#) functionality to streamline remote access and improve user experiences.

Classification of Assets

Assets are generally classified in three ways:

1. **Convertibility:** Classifying assets based on how easy it is to convert them into cash.
2. **Physical Existence:** Classifying assets based on their physical existence (in other words, tangible vs. intangible assets).
3. **Usage:** Classifying assets based on their business operation usage/purpose.



Physical Security Assessment

When people think about [physical security](#), they often imagine a simple process. Lock the doors, make sure smoke detectors and alarms are working, and mount cameras around entrances to the building.

Covering the basics is important, but the devil is in the details. The door to the loading dock doesn't quite latch closed sometimes. There's a light in the parking lot stairwell that's been out for two weeks. You haven't updated evacuation plans and maps after a company reorganization.

None of these issues is an immediate hazard, but they all represent potential security threats that could harm your company's people and property. As a security professional, mitigating these identified risks before they materialize is one of your core objectives. But to solve a problem, you need to know it exists in the first place. That's where a physical security assessment comes in.

security Assets:

In the realm of cybersecurity, **an asset is anything that holds value to an organization** and can be affected by cyber threats. These assets span a wide range of elements, from physical devices to information and people. The scope of cybersecurity assets is broad and includes:

- **Hardware:** Servers, computers, and mobile devices
- **Software:** Applications, operating systems, and databases
- **Data:** Sensitive information, customer details, intellectual property
- **Networks:** Infrastructure, architecture, and internet access
- **People:** Employees, customers, and partners who have access to technological resources

Each asset has a role in the operational integrity of a company and a potential point of vulnerability. It's imperative for organizations to accurately identify and classify their assets to implement appropriate security measures. This process begins with an asset inventory, which is a comprehensive list detailing all the elements that need protection.

Lock and entry control in security

A door access system is an advanced security solution designed to manage and monitor who can enter or exit specific locations within a facility.

Unlike traditional keys, which can be copied or lost, these systems utilize digital mechanisms such as key cards, biometrics, or PIN codes to control access [1].

By incorporating technology, access control systems offer a more secure, flexible, and convenient way to ensure that only authorized individuals can access certain areas, keeping intruders and potential threats at bay.

How Does A Door Access Control Work?

At its core, a door access control solution operates using a combination of [electronic door locks](#), an access controller (a card reader, fingerprint scanner, or keypad), and a control panel or cloud-based system that manages the users' access permissions.

When an individual attempts entry, the reader scans the provided credentials, whether a card, fingerprint, or code. The system then checks these against its database.

Access is granted if the credentials match an authorized user; otherwise, the door remains locked. Additionally, all access events are typically logged, providing valuable insights into the flow of personnel and enhancing the access control security infrastructure.

Physical Intrusion Detection Systems

Controlling access to physical and digital spaces using authentication, credentials, and other measures is just one part of [access control](#). Especially for larger organizations, with larger security footprints, the ability to spot potential threats and send alerts is critical in preventing, or at least mitigating, breaches.

What Are Physical Intrusion Detection Systems (PIDS)?

To achieve this level of security, many organizations deploy an intrusion detection system (IDS) or intrusion prevention system of some form. An IDS is an application or physical appliance that monitors access points for suspicious activity that might indicate an intrusion; an intrusion prevention system (IPS) shuts down a security breach as soon as it's discovered (which may include initiating a building lockdown.) The main difference between them is that IDS is an access monitoring system, while IPS is an access control system.